

6. Geometry

Ref	Content	Notes
6.1		Knowledge of perimeter and area of rectangles and circles; and of the area of triangles, parallelograms and trapezia; and of the surface area and volume of prisms, cylinders, spheres, cones and pyramids Knowledge of angle properties of parallel and intersecting lines, triangles, all special types of quadrilaterals and polygons Understand and use circle theorems: Angle at the centre is twice the angle at the circumference; angles in the same segment are equal; opposite angles in cyclic quadrilateral add up to 180°; alternate segment theorem; the theorems listed in the notes of section 3.7

Geometric proof

6.2	Understand and construct geometrical proofs using formal arguments	The use of theorems listed in the notes of 3.7 and 6.1
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Trigonometry in triangles

6.3	Sine and cosine rules in scalene triangles; area of a triangle $= \frac{1}{2}ab \sin C$	Knowledge and use of trigonometry to solve right-angled triangles is expected
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Pythagoras' theorem

6.4	Use of Pythagoras' theorem in 2D and 3D	Recognise Pythagorean triples; 3, 4, 5; 5, 12, 13; 8, 15, 17; 7, 24, 25 and simple multiples of these
6.5	Be able to apply trigonometry and Pythagoras' theorem to 2 and 3 dimensional problems	Including the angle between a line and a plane and the angle between two planes; including triangles that do not have right angles

Ratios of angles and their graphs

6.6	Sketch and use graphs of $y = \sin x$, $y = \cos x$ and $y = \tan x$ for angles of any size	
6.7	Be able to use the definitions $\sin \theta$, $\cos \theta$ and $\tan \theta$, for any positive angle up to 360° (measured in degrees only)	Angles measured anticlockwise will be taken as positive
6.8	Knowledge and use of 30° , 60° , 90° triangles and 45° , 45° , 90° triangles	The use of the ratios $1 : \sqrt{3} : 2$ and $1 : 1 : \sqrt{2}$

Ref	Content	Notes
6.9	Know and use $\tan \theta = \frac{\sin \theta}{\cos \theta}$ and $\sin^2 \theta + \cos^2 \theta = 1$	Including expressions to be simplified, proofs of identities and equations solved
6.10	Solution of simple trigonometric equations in given intervals	Equations will be restricted to single angles: $\sin x = 0.5$; $\sqrt{2} \sin x = \cos x$ for $0^\circ \leq x \leq 360^\circ$; $\sin^2 x = \frac{1}{4}$ for $0^\circ \leq x \leq 360^\circ$